Flow: Fuzzy Guidance for Dynamic Electricity Tariffs

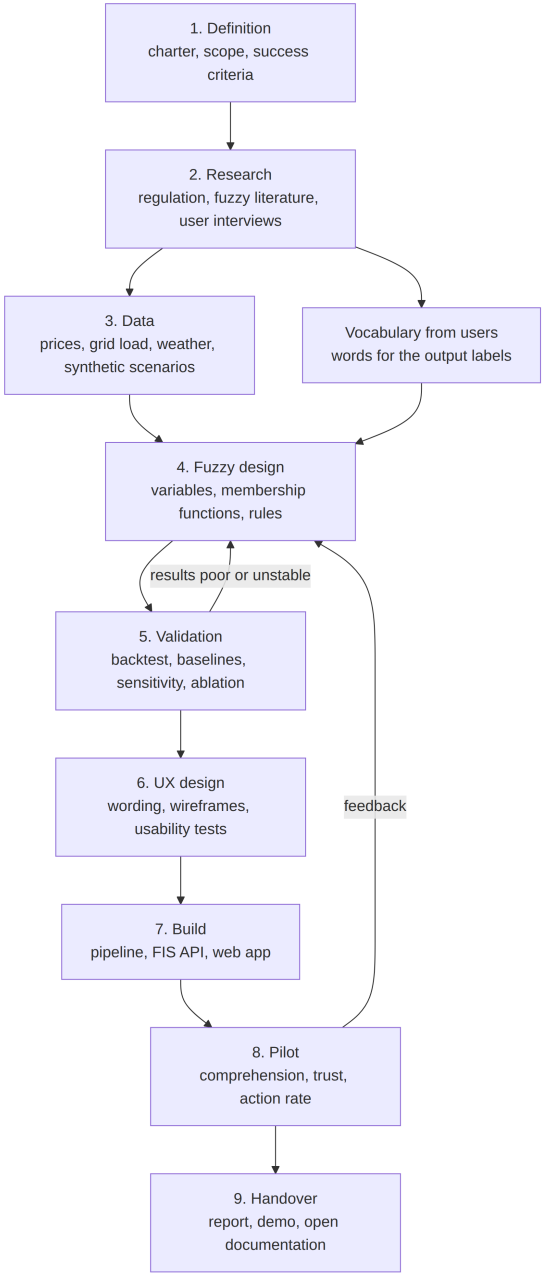
This file shows how the project and the fuzzy system fit together. Diagrams use Mermaid and render on GitHub, VS Code (with a Mermaid extension) and most Markdown viewers.

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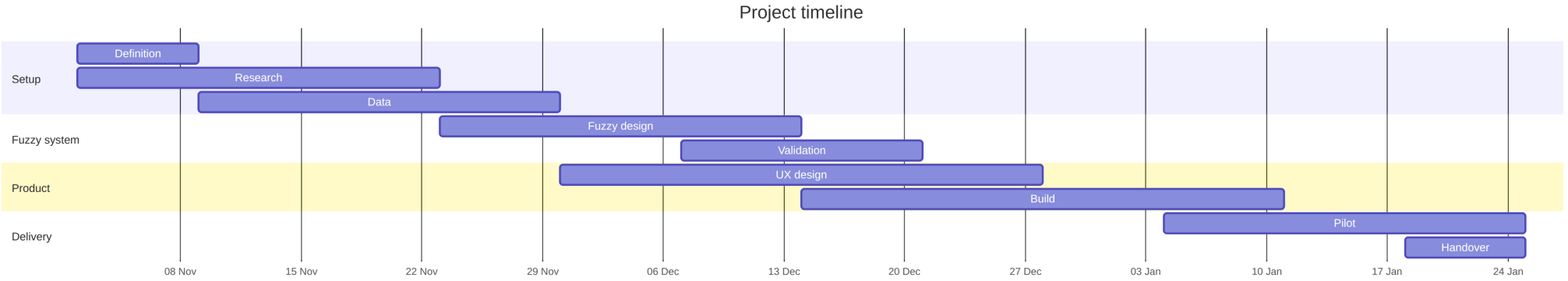
1. Project workflow end to end
2. Timeline
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4. Two-stage structure (price label, then device advisor)
5. Worked example
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1. Project workflow end to end

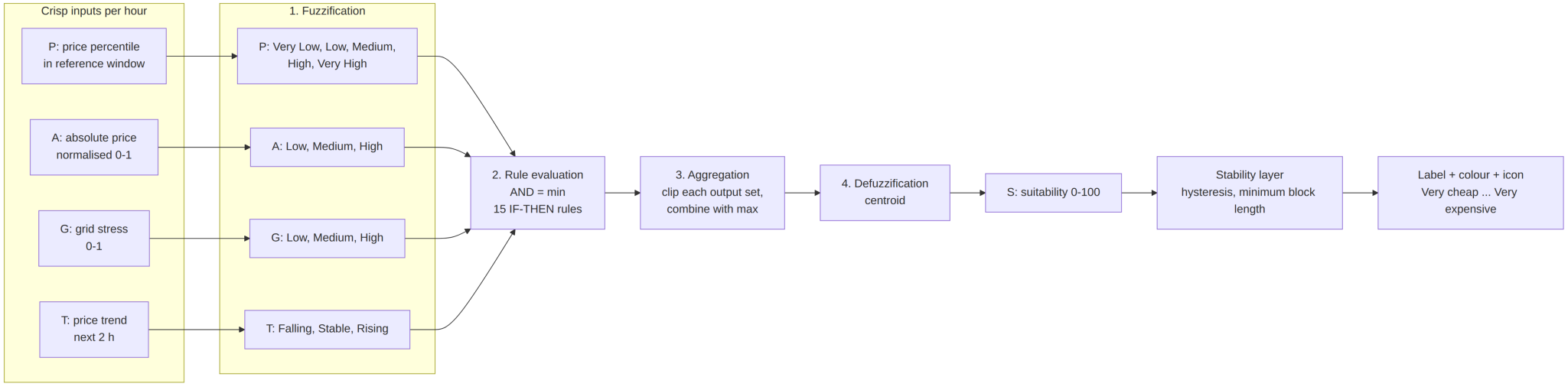


2. Timeline (12 weeks, example start 2 Nov 2026)



Adjust dates for holidays and your actual deadline.

3. Fuzzy inference pipeline (Mamdani)

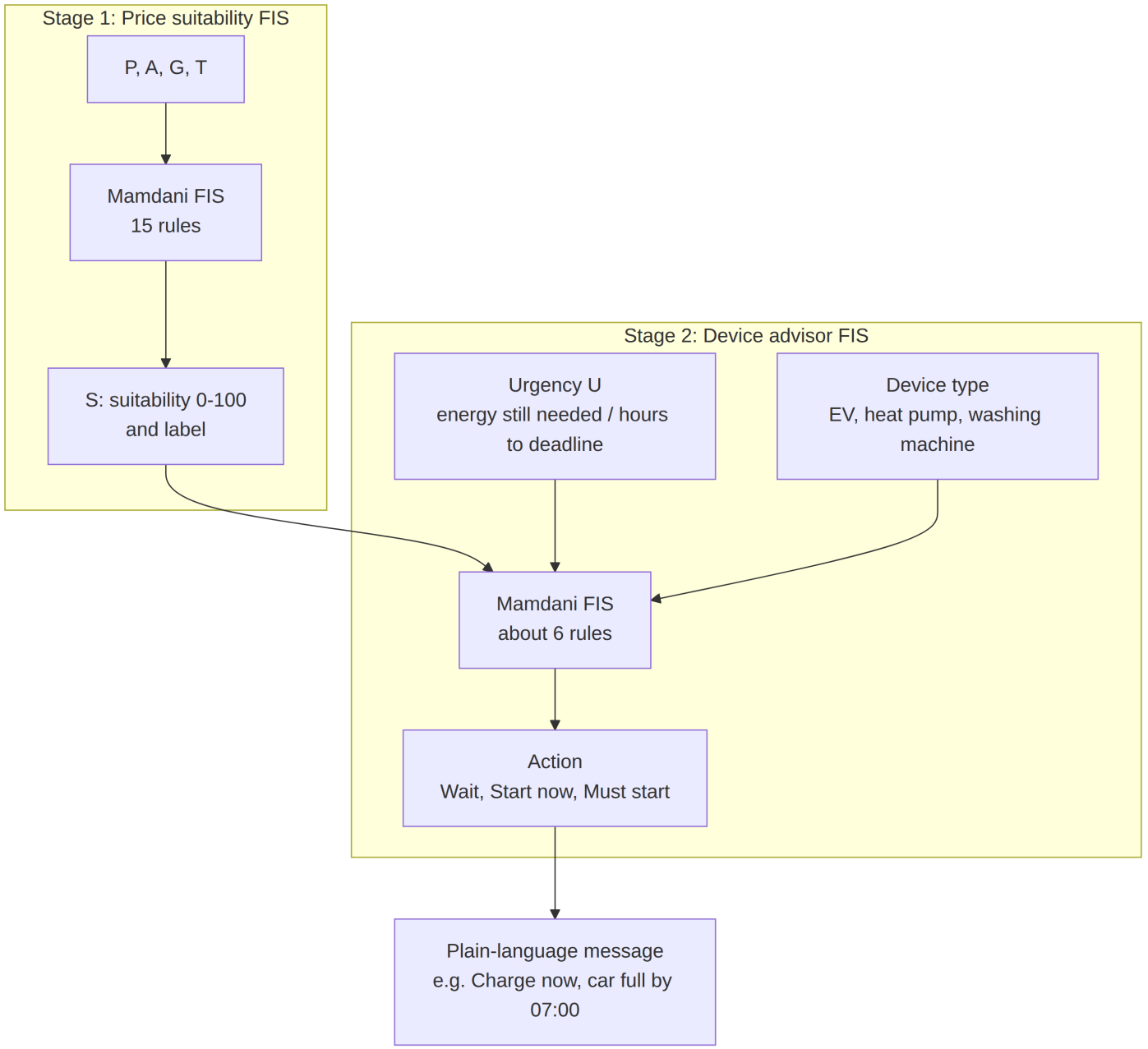


Membership functions (starting values, to be calibrated on data)

Variable	Set	Shape	Parameters
P (0-100)	Very Low	trapezoid	0, 0, 10, 25
P	Low	triangle	10, 25, 45
P	Medium	triangle	35, 50, 65
P	High	triangle	55, 75, 90
P	Very High	trapezoid	75, 90, 100, 100
A (0-1)	Low	trapezoid	0, 0, 0.25, 0.5
A	Medium	triangle	0.25, 0.5, 0.75
A	High	trapezoid	0.5, 0.75, 1, 1
G (0-1)	Low, Medium, High	same shapes as A	
S (0-100)	Avoid	trapezoid	0, 0, 10, 25
S	Wait	triangle	10, 30, 50
S	Okay	triangle	40, 55, 70
S	Good	triangle	60, 75, 90
S	Ideal	trapezoid	75, 90, 100, 100

4. Two-stage structure

Stage 1 judges the hour. Stage 2 adds the user's situation. Splitting the system keeps each rule base small and explainable.


Stage 1: Price suitability FIS
P, A, G, T
Mamdani FIS
15 rules
S: suitability 0-100
and label
Stage 2: Device advisor FIS
Urgency U
energy still needed / hours
to deadline
Device type
EV, heat pump, washing
machine
Mamdani FIS
about 6 rules
Action
Wait, Start now, Must start
Plain-language message
e.g. Charge now, car full by
07:00

5. Worked example

One hour on a winter night. The numbers below were calculated by hand and should be confirmed by the code.

Crisp inputs: P = 12, A = 0.30, G = 0.20, T = stable

Step 1: Fuzzification

Variable	Membership degrees
P = 12	Very Low 0.87, Low 0.13
A = 0.30	Low 0.80, Medium 0.20
G = 0.20	Low 1.0
T = stable	Stable 1.0

Step 2: Rules that fire (AND = min)

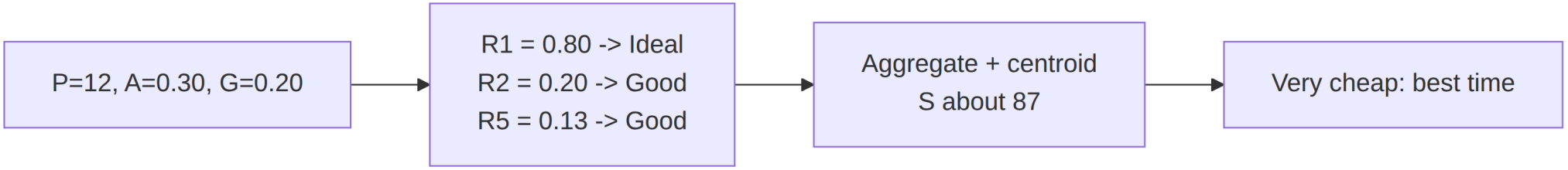
Rule	Calculation	Strength	Output set
R1: P Very Low AND A Low	min(0.87, 0.80)	0.80	Ideal
R2: P Very Low AND A Medium	min(0.87, 0.20)	0.20	Good
R5: P Low AND G Low	min(0.13, 1.0)	0.13	Good

All other rules have strength 0.

Step 3: Aggregation. Ideal is clipped at 0.80. Good gets max(0.20, 0.13) = 0.20 and is clipped there. The two clipped shapes are combined with max.

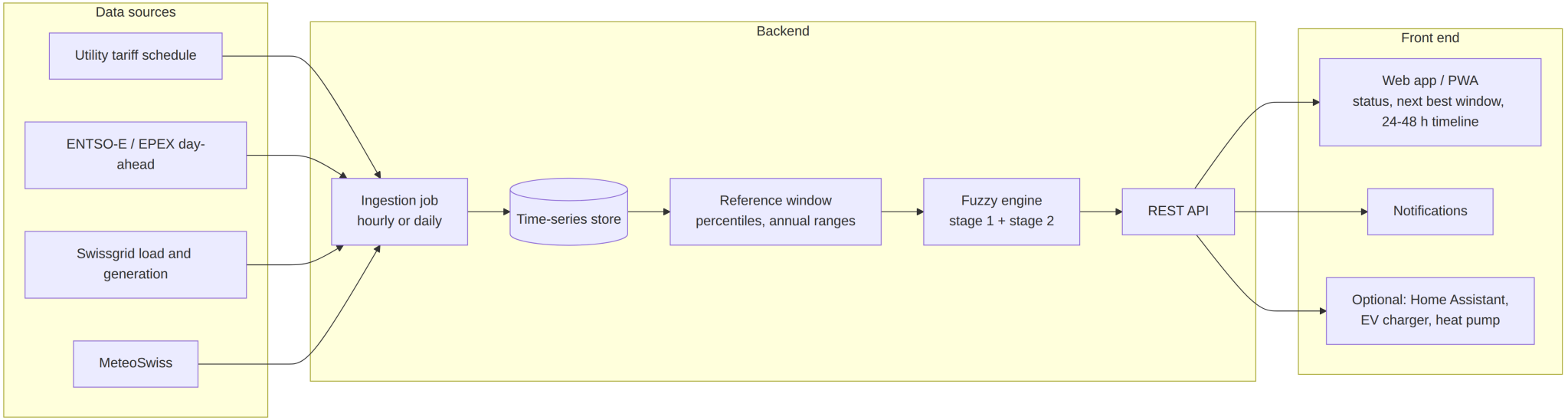
Step 4: Defuzzification. The centroid of the combined shape is about **S = 87**.

Result: label **"Very cheap: best time"**, and the device advisor can say "Charge the car now" if the user needs the energy by the morning.

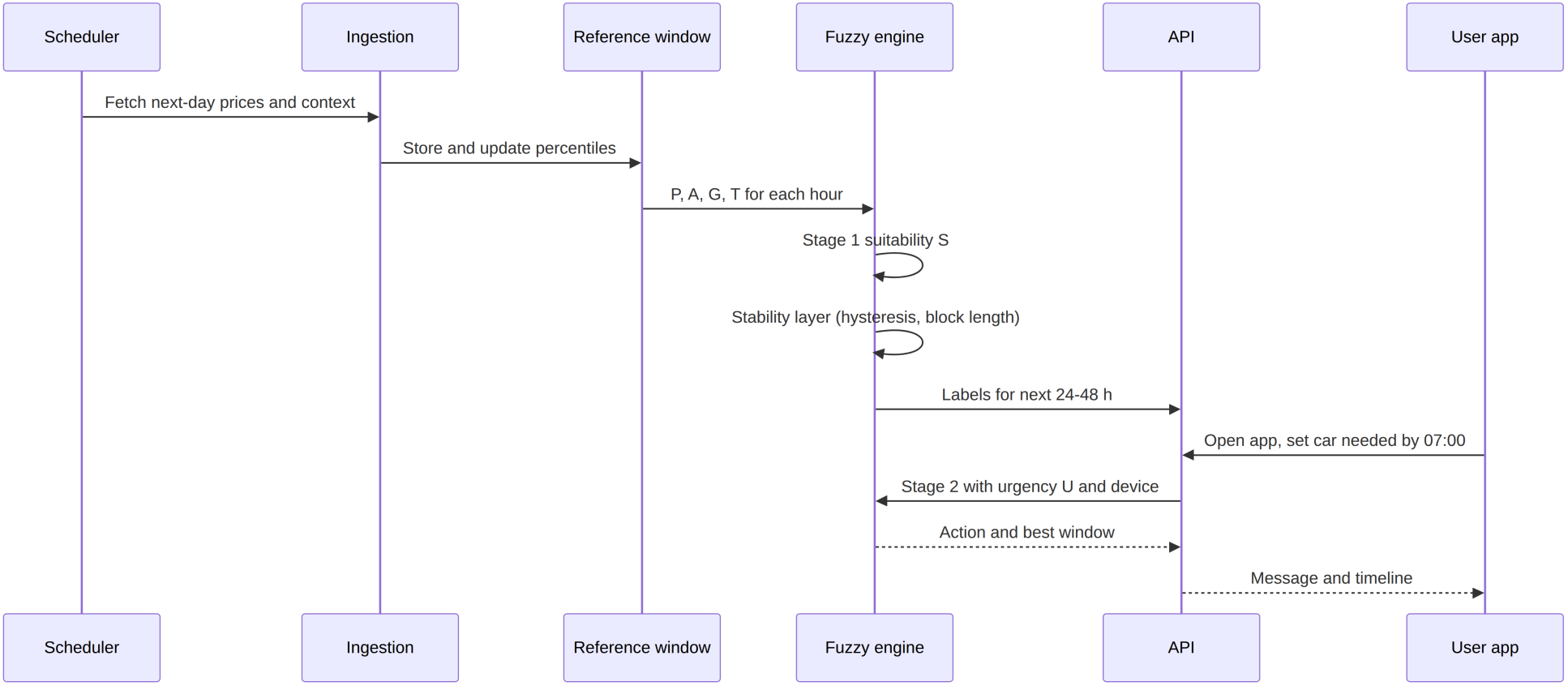


6. System architecture and runtime sequence

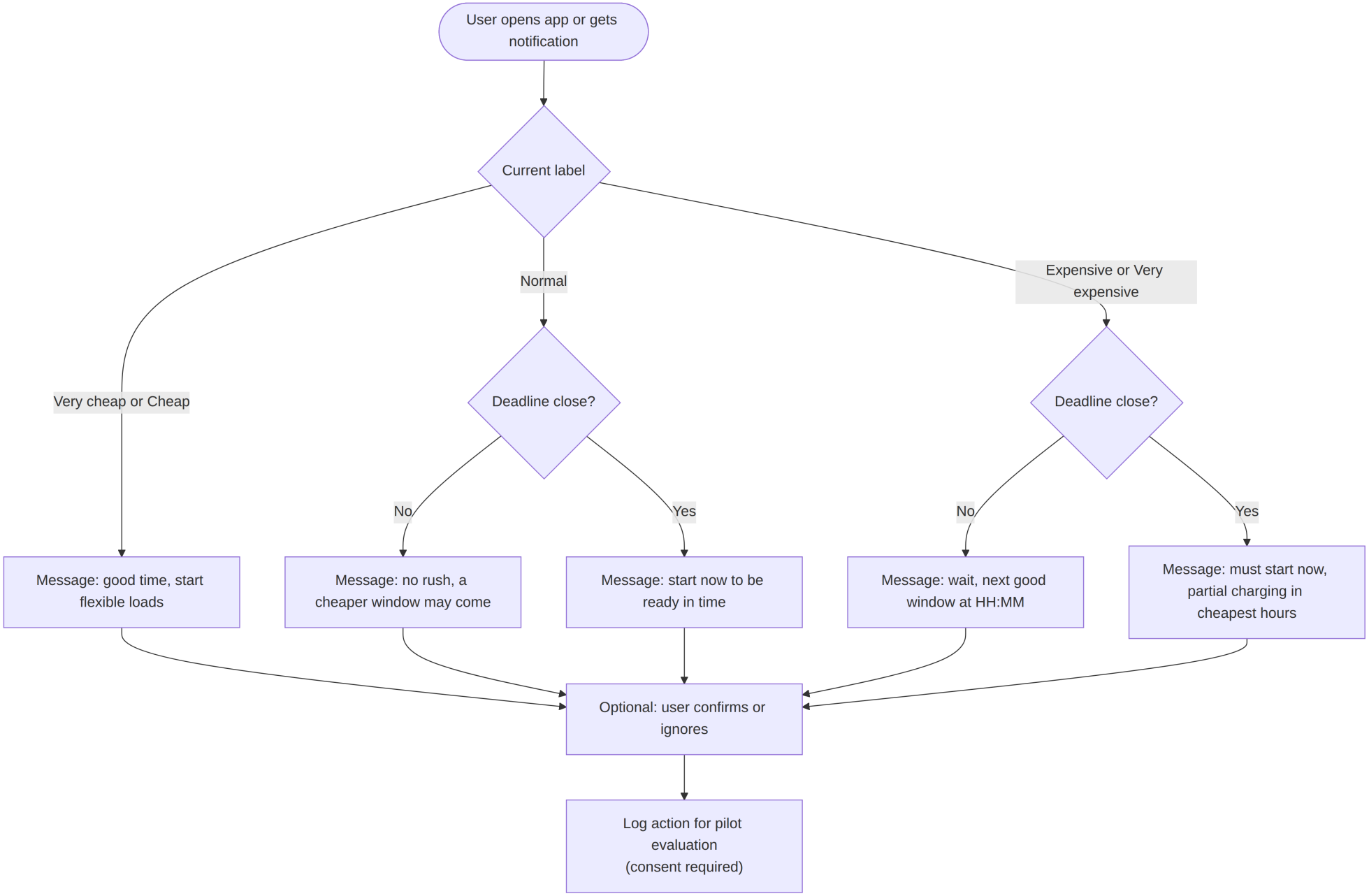
Architecture



Runtime sequence



7. User decision flow



Evaluation flow

